



National Accreditation Board for Testing and Calibration Laboratories

SCOPE OF ACCREDITATION

Laboratory Name :

BHARTI AUTOMATION PRIVATE LIMITED, PLOT NO. 354, SECTOR - 7, IMT
MANESAR, GURGAON, HARYANA, INDIA

Accreditation Standard

ISO/IEC 17025:2017

Certificate Number

CC-3068

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Validity

23/10/2024 to 22/10/2028

Last Amended on

20/01/2025

S.No	Discipline / Group	Measurand or Reference Material/Type of instrument or material to be calibrated or measured / Quantity Measured /Instrument	Calibration or Measurement Method or procedure	Measurement range and additional parameters where applicable(Range and Frequency)	* Calibration and Measurement Capability(CMC)(±)
Permanent Facility					
1	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Current @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 A to 10 A	0.39 % to 0.18 %
2	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Current @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 mA to 100 mA	0.76 % to 0.57 %
3	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Current @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	100 mA to 1 A	0.57 % to 0.39 %
4	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Resistance @ 1 kHz - 2 Wire	Using LCR Meter by Direct Method	1 ohm to 100 kohm	0.15 % to 0.08 %
5	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Voltage @ 50 Hz	Using Digital Multimeter with High Voltage Probe by Direct Method	1 kV to 10 kV	3 %
6	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Voltage @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 V to 750 V	0.86 %



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7	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Voltage @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	10 mV to 100 mV	0.55 % to 0.63 %
8	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Voltage @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	100 mV to 1 V	0.63 % to 0.86 %
9	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	Capacitance @ 1 kHz	Using LCR Meter by Direct Method	100 pF to 100 µF	0.55 % to 0.56 %
10	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	Inductance @ 1 kHz	Using LCR Meter by Direct Method	100 µH to 10 H	0.68 % to 0.82 %
11	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Source)	1Ø, 4 Wire, AC Active Energy @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.005 kWh to 0.9 kWh	0.25 %
12	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Source)	1Ø, 4 Wire, AC Active Power @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.01 kW to 1.8 kW	0.27 %



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13	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	1Ø, 4 Wire, AC Reactive Energy @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.008 kVarh to 0.783 kVarh	0.25 %
14	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	1Ø, 4 Wire, AC Reactive Power @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.017 kVAr to 1.558 kVAr	0.27 %
15	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	3Ø, 4 Wire, AC Active Energy @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.015 kWh to 2.7 kWh	0.25 %
16	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	3Ø, 4 Wire, AC Active Power @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.03 kW to 5.4 kW	0.27 %
17	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	3Ø, 4 Wire, AC Reactive Energy @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.024 kVarh to 2.349 kVarh	0.25 %
18	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	3Ø, 4 Wire, AC Reactive Power @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.052 kVAr to 4.677 kVAr	0.27 %
19	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator by Direct Method	1 A to 10 A	0.5 % to 0.48 %



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20	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator by Direct Method	1 mA to 100 mA	1 % to 0.4 %
21	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator with 100 Turns Current Coil by Direct Method	10 A to 100 A	0.48 % to 2 %
22	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator by Direct Method	100 mA to 1 A	0.4 % to 0.5 %
23	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator by Direct Method	100 µA to 1 mA	0.79 % to 1 %
24	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator with Current Coil by Direct Method	100 A to 1000 A	2 % to 0.8 %
25	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Voltage @ 50 Hz	Using Multi Function Calibrator by Direct Method	100 V to 1000 V	0.27 % to 0.36 %
26	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Voltage @ 50 Hz	Using Multi Function Calibrator by Direct Method	20 mV to 100 V	1.08 % to 0.27 %



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27	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Capacitance @ 1 kHz	Using Capacitance Box by Direct Method	1 nF to 10 μ F	0.46 % to 0.16 %
28	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Inductance @ 1 kHz	Using Inductance Box by Direct Method	1 mH to 10 mH	0.73 % to 0.84 %
29	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Inductance @ 1 kHz	Using Inductance Box by Direct Method	10 mH to 1 H	0.84 % to 0.61 %
30	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Power Factor @ 50 Hz	Using Standard 3 $\emptyset$ Energy Source by Direct Method	0.1 (Lead / Lag) to UPF	0.008 PF
31	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Resistance @ 1 kHz - 2 Wire	Using Decade Resistance Box by Direct Method	1 ohm to 100 kohm	0.5 % to 0.58 %
32	ELECTRO-TECHNICAL- DIRECT CURRENT (Measure)	Resistance - 4 Wire	Using Digital Micro Ohm Meter by Direct Method	20 μ ohm to 20 kohm	0.9 % to 0.65 %
33	ELECTRO-TECHNICAL- DIRECT CURRENT (Measure)	DC Current	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 A to 10 A	0.12 % to 0.19 %



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34	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Current	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 mA to 1 A	0.76 % to 0.12 %
35	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC High Voltage	Using Digital Multimeter with High Voltage Probe by Direct Method	1 kV to 10 kV	2.4 %
36	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	1 kohm to 10 kohm	0.37 % to 0.13 %
37	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	1 Mohm to 10 Mohm	0.13 % to 0.05 %
38	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	10 kohm to 1 Mohm	0.13 %
39	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	10 Mohm to 100 Mohm	0.05 % to 1.51 %
40	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	10 ohm to 100 ohm	0.82 % to 0.13 %



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41	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	100 ohm to 1 kohm	0.13 % to 0.37 %
42	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Voltage	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 V to 100 V	0.18 % to 0.11 %
43	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Voltage	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	10 mV to 100 mV	0.71 % to 0.18 %
44	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Voltage	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	100 mV to 1 V	0.18 %
45	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Voltage	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	100 V to 1000 V	0.11 % to 0.1 %
46	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator by Direct Method	1 A to 10 A	0.5 %
47	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator by Direct Method	1 mA to 100 mA	0.3 %



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48	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator with 100 Turn Current Coil by Direct Method	10 A to 100 A	0.5 % to 2 %
49	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator with 100 Turn Current Coil by Direct Method	100 A to 1000 A	2 % to 0.95 %
50	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator by Direct Method	100 mA to 1 A	0.3 % to 0.5 %
51	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Voltage	Using Multi Function Calibrator by Direct Method	10 mV to 100 mV	0.31 %
52	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Voltage	Using Multi Function Calibrator by Direct Method	10 V to 100 V	0.21 % to 0.19 %
53	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Voltage	Using Multi Function Calibrator by Direct Method	100 mV to 10 V	0.31 % to 0.21 %
54	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Voltage	Using Multi Function Calibrator by Direct Method	100 V to 1000 V	0.19 %



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55	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box (Discrete Values) by Direct Method	0.001 ohm to 1 kohm	1.61 %
56	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box by Direct Method	1 kohm to 5 kohm	1.61 %
57	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Mega Ohm Box by Direct Method	1 Mohm to 5 Mohm	0.8 %
58	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box (Discrete Values) by Direct Method	100 kohm to 1000 kohm	0.13 % to 0.8 %
59	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box by Direct Method	5 kohm to 10 kohm	1.61 % to 0.12 %
60	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box (Discrete Values) by Direct Method	5 Mohm to 100 Mohm	0.8 % to 1.5 %
61	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box by Direct Method	50 kohm to 100 kohm	0.12 % to 0.13 %



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62	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using Resistance Box & Decade Resistance Box (Discrete Values) by Direct Method	100 Mohm to 20 Gohm	1.5 % to 2.75 %
63	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box by Direct Method	10 kohm to 50 kohm	0.12 %
64	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using HV Mega Ohm Box (Discrete Values) by Direct Method	2 Gohm to 20 Gohm	2.25 % to 2.75 %
65	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using HV Mega Ohm Box (Discrete Values) by Direct Method	20 Mohm to 200 Mohm	1.5 % to 1.75 %
66	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using HV Mega Ohm Box (Discrete Values) by Direct Method	200 Mohm to 2 Gohm	1.75 % to 2.25 %
67	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using HV Mega Ohm Box (Discrete Values) by Direct Method	2 Mohm to 20 Mohm	0.8 % to 1.5 %
68	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 4 Wire	Using Standard Resistance Box (Discrete Values) by Direct Method	20 µohm to 100 kohm	0.47 % to 0.12 %



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69	ELECTRO-TECHNICAL-ELECTRICAL EQUIPMENT (Measure)	AC Energy 3 $\emptyset$, 4 Wire @ (50 Hz, UPF to 0.5 lag, 240 V, 1.25 A to 5 A)	Using Standard Energy Meter by Direct Method	0.45 kWh to 3.6 kWh	0.38 %
70	ELECTRO-TECHNICAL-ELECTRICAL EQUIPMENT (Measure)	AC Power, 3 $\emptyset$, 4 Wire @ (50 Hz, UPF to 0.5 lag, 240 V, 1.25 A to 5 A)	Using Standard Energy Meter by Direct Method	0.45 kW to 3.6 kW	0.36 %
71	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	RTD (PT 100)	Using Universal Calibrator by Direct Method	(-) 100 °C to 650 °C	0.83 °C
72	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	Thermocouple B Type	Using Universal Calibrator by Direct Method	0 °C to 600 °C	0.82 °C
73	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	Thermocouple E Type	Using Universal Calibrator by Direct Method	(-) 100 °C to 1000 °C	0.91 °C
74	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	Thermocouple J Type	Using Universal Calibrator by Direct Method	(-) 100 °C to 700 °C	0.83 °C
75	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	Thermocouple K Type	Using Universal Calibrator by Direct Method	(-) 200 °C to 1300 °C	0.83 °C



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76	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Measure)	Thermocouple T Type	Using Universal Calibrator by Direct Method	0 °C to 400 °C	0.92 °C
77	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	RTD (PT 100)	Using Universal Calibrator by Direct Method	(-) 100 °C to 650 °C	0.96 °C
78	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple B Type	Using Universal Calibrator by Direct Method	0 °C to 600 °C	0.59 °C
79	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple E Type	Using Universal Calibrator by Direct Method	(-) 100 °C to 1000 °C	0.83 °C
80	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple J Type	Using Universal Calibrator by Direct Method	(-) 200 °C to 700 °C	0.83 °C
81	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple K Type	Using Universal Calibrator by Direct Method	(-) 200 °C to 1300 °C	0.84 °C
82	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple T Type	Using Universal Calibrator by Direct Method	0 °C to 400 °C	0.83 °C



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83	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Frequency	Using 6½ Digit Multimeter by Direct Method	10 Hz to 100 Hz	0.58 % to 0.06 %
84	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Frequency	Using 6½ Digit Multimeter by Direct Method	100 Hz to 100 kHz	0.06 % to 0.03 %
85	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Frequency	Using 6½ Digit Multimeter by Direct Method	100 kHz to 1000 kHz	0.03 %
86	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Time	Using Time Interval Meter by Comparison Method	1 hour to 8 hour	0.44 s to 11.8 s
87	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Time	Using Time Interval Meter by Comparison Method	1 minute to 1 hour	0.017 s to 0.44 s
88	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Time	Using Time Interval Meter by Comparison Method	10 s to 1 minute	1.9 ms to 0.017 s
89	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Time	Using Time Interval Meter by Comparison Method	8 hour to 24 hour	11.8 s to 23 s



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90	FLUID FLOW-FLOW MEASURING DEVICES	Dial Flow Meter, Digital Flow Meter, Flow Transmitter, Flow Transducer (Medium : Water)	Using Ultrasonic Flow Meter by Comparison Method	> 100 m ³ /h to 250 m ³ /h	1.09 %
91	FLUID FLOW-FLOW MEASURING DEVICES	Dial Flow Meter, Digital Flow Meter, Flow Transmitter, Flow Transducer (Medium : Water)	Using Ultrasonic Flow Meter by Comparison Method	1 m ³ /h to 100 m ³ /h	1.09 %
92	FLUID FLOW-FLOW MEASURING DEVICES	Magnetic Flow Meter, Liquid Turbine Flow Meter, Ultrasonic Flow Meter (Medium : Water)	Using 150 kg Weighing System by Gravimetric Method	0.01 m ³ to 0.065 m ³	0.24 %
93	FLUID FLOW-FLOW MEASURING DEVICES	Magnetic Flow Meter, Liquid Turbine Flow Meter, Ultrasonic Flow Meter (Medium : Water)	Using 600 kg Weighing System by Gravimetric Method	0.1 m ³ to 0.375 m ³	0.23 %
94	FLUID FLOW-FLOW MEASURING DEVICES	Magnetic Flow Meter, Liquid Turbine Flow Meter, Ultrasonic Flow Meter (Medium : Water)	Using 150 kg Weighing System by Gravimetric Method	0.5 m ³ /h to 2.5 m ³ /h	0.29 %
95	FLUID FLOW-FLOW MEASURING DEVICES	Magnetic Flow Meter, Liquid Turbine Flow Meter, Ultrasonic Flow Meter (Medium : Water)	Using 3000 kg Weighing System by Gravimetric Method	0.51 m ³ to 1.008 m ³	0.24 %



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96	FLUID FLOW-FLOW MEASURING DEVICES	Magnetic Flow Meter, Liquid Turbine Flow Meter, Ultrasonic Flow Meter (Medium : Water)	Using 3000 kg Weighing System by Gravimetric Method	25 m ³ /h to 250 m ³ /h	0.26 %
97	FLUID FLOW-FLOW MEASURING DEVICES	Magnetic Flow Meter, Liquid Turbine Flow Meter, Ultrasonic Flow Meter (Medium : Water)	Using 600 kg Weighing System by Gravimetric Method	5 m ³ /h to 25 m ³ /h	0.28 %
98	FLUID FLOW-FLOW MEASURING DEVICES	Mass Flow Meter, Coriolis Flow Meter (Medium : Water)	Using 600 kg Weighing System by Gravimetric Method	100 kg to 370 kg	0.2 %
99	FLUID FLOW-FLOW MEASURING DEVICES	Mass Flow Meter, Coriolis Flow Meter (Medium : Water)	Using 3000 kg Weighing System by Gravimetric Method	25 TPH to 120 TPH	0.26 %
100	FLUID FLOW-FLOW MEASURING DEVICES	Mass Flow Meter, Coriolis Flow Meter (Medium : Water)	Using 150 kg Weighing System by Gravimetric Method	0.5 TPH to 2.5 TPH	0.29 %
101	FLUID FLOW-FLOW MEASURING DEVICES	Mass Flow Meter, Coriolis Flow Meter (Medium : Water)	Using 150 kg Weighing System by Gravimetric Method	10 kg to 62 kg	0.23 %
102	FLUID FLOW-FLOW MEASURING DEVICES	Mass Flow Meter, Coriolis Flow Meter (Medium : Water)	Using 600 kg Weighing System by Gravimetric Method	5 TPH to 22 TPH	0.27 %
103	FLUID FLOW-FLOW MEASURING DEVICES	Mass Flow Meter, Coriolis Flow Meter (Medium : Water)	Using 3000 kg Weighing System by Gravimetric Method	508 kg to 1005 kg	0.24 %



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104	MECHANICAL-ACCELERATION AND SPEED	Centrifuge - Contact Type	Using Digital Tachometer by Comparison Method	> 1000 rpm to 5000 rpm	4 rpm
105	MECHANICAL-ACCELERATION AND SPEED	Centrifuge - Contact Type	Using Digital Tachometer by Comparison Method	10 rpm to 1000 rpm	1.7 rpm
106	MECHANICAL-ACCELERATION AND SPEED	Centrifuge - Non Contact Type	Using Digital Tachometer by Comparison Method	> 50000 rpm to 99000 rpm	58.03 rpm
107	MECHANICAL-ACCELERATION AND SPEED	Centrifuge - Non Contact Type	Using Digital Tachometer by Comparison Method	> 100 rpm to 10000 rpm	7.06 rpm
108	MECHANICAL-ACCELERATION AND SPEED	Centrifuge - Non Contact Type	Using Digital Tachometer by Comparison Method	> 10000 rpm to 50000 rpm	30.64 rpm
109	MECHANICAL-ACCELERATION AND SPEED	Centrifuge - Non Contact Type	Using Digital Tachometer by Comparison Method	10 rpm to 100 rpm	1 rpm
110	MECHANICAL-ACCELERATION AND SPEED	Stroboscope, Tachometer, Pulse Engine Tachometer - Non Contact Type	Digital Tachometer and RPM Source by Comparison Method	> 100 rpm to 10000 rpm	7.73 rpm
111	MECHANICAL-ACCELERATION AND SPEED	Stroboscope, Tachometer, Pulse Engine Tachometer - Non Contact Type	Using Digital Tachometer and RPM Source by Comparison Method	> 10000 rpm to 50000 rpm	30.67 rpm
112	MECHANICAL-ACCELERATION AND SPEED	Stroboscope, Tachometer, Pulse Engine Tachometer - Non Contact Type	Using Digital Tachometer and RPM Source by Comparison Method	> 50000 rpm to 99000 rpm	58.02 rpm
113	MECHANICAL-ACCELERATION AND SPEED	Stroboscope, Tachometer, Pulse Engine Tachometer - Non Contact Type	Using Digital Tachometer and RPM Source by Comparison Method	10 rpm to 100 rpm	1 rpm



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114	MECHANICAL-ACCELERATION AND SPEED	Tachometer - Contact Type	Using Digital Tachometer and RPM Source by Comparison Method	> 1000 rpm to 5000 rpm	7.63 rpm
115	MECHANICAL-ACCELERATION AND SPEED	Tachometer - Contact Type	Using Digital Tachometer and RPM Source by Comparison Method	10 rpm to 1000 rpm	2.26 rpm
116	MECHANICAL-ACCELERATION AND SPEED	Tachometer - Non Contact Type	Using Digital Tachometer and RPM Source by Comparison Method	> 10000 rpm to 50000 rpm	36.91 rpm
117	MECHANICAL-ACCELERATION AND SPEED	Tachometer - Non Contact Type	Using Digital Tachometer and RPM Source by Comparison Method	> 100 rpm to 10000 rpm	24.46 rpm
118	MECHANICAL-ACCELERATION AND SPEED	Tachometer - Non Contact Type	Using Digital Tachometer and RPM Source by Comparison Method	> 50000 rpm to 99000 rpm	61.69 rpm
119	MECHANICAL-ACCELERATION AND SPEED	Tachometer - Non Contact Type	Using Digital Tachometer and RPM Source by Comparison Method	10 rpm to 100 rpm	2.1 rpm
120	MECHANICAL-ACCELERATION AND SPEED	Vibration Meter - Acceleration @ 10 Hz (pk)	Using Vibration Meter, Vibration Test System by Comparison Method	3.99 m/s ²	4.9 %
121	MECHANICAL-ACCELERATION AND SPEED	Vibration Meter - Acceleration @ 160 Hz (pk)	Using Vibration Meter, Vibration Test System by Comparison Method	50.36 m/s ² to 249.32 m/s ²	4.9 %
122	MECHANICAL-ACCELERATION AND SPEED	Vibration Meter - Acceleration @ 80 Hz (pk)	Using Vibration Meter, Vibration Test System by Comparison Method	25.16 m/s ² to 100.82 m/s ²	4.9 %



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123	MECHANICAL- ACCELERATION AND SPEED	Vibration Meter - Displacement @ 10 Hz (pk-pk)	Using Vibration Meter, Vibration Test System by Comparison Method	0.5 mm to 2 mm	10.2 %
124	MECHANICAL- ACCELERATION AND SPEED	Vibration Meter - Displacement @ 160 Hz (pk-pk)	Using Vibration Meter, Vibration Test System by Comparison Method	0.027 mm to 0.5 mm	13.17 %
125	MECHANICAL- ACCELERATION AND SPEED	Vibration Meter - Displacement @ 80 Hz (pk-pk)	Using Vibration Meter, Vibration Test System by Comparison Method	0.2 mm to 0.81 mm	13.17 %
126	MECHANICAL- ACCELERATION AND SPEED	Vibration Meter - Velocity @ 10 Hz (RMS)	Using Vibration Meter, Vibration Test System by Comparison Method	1.07 mm/s to 60.88 mm/s	4.83 %
127	MECHANICAL- ACCELERATION AND SPEED	Vibration Meter - Velocity @ 160 Hz (RMS)	Using Vibration Meter, Vibration Test System by Comparison Method	35.11 mm/s to 104.82 mm/s	4.83 %
128	MECHANICAL- ACCELERATION AND SPEED	Vibration Meter - Velocity @ 80 Hz (RMS)	Using Vibration Meter, Vibration Test System by Comparison Method	50.02 mm/s to 141.03 mm/s	4.83 %
129	MECHANICAL- ACOUSTICS	Sound Level Meter @ 1 kHz	Using Sound Level Calibrator by Direct Method	114 dB	0.71 dB
130	MECHANICAL- ACOUSTICS	Sound Level Meter @ 1 kHz	Using Sound Level Calibrator by Direct Method	94 dB	0.71 dB



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131	MECHANICAL-DENSITY AND VISCOSITY	Density Hydrometer	Using Electronic Weighing Balance (Readability : 0.1 mg) and Standard Hydrometers based on IS 3104:1982 (Reaffirmed Year : 2018)	0.65 g/ml to 1.85 g/ml	0.007 g/ml
132	MECHANICAL-DENSITY AND VISCOSITY	Lactometer	Using Standard Hydrometers and Liquid of known Densities as per IS 9585:1980 (Reaffirmed Year : 2018)	0.65 Sp. Gr to 1.85 Sp. Gr	0.008 Sp. Gr
133	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Bevel Protractor (L.C.: 5 minute of arc)	Using Profile Projector by Comparison Method	0 ° to 360 °	3 minute of arc
134	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Caliper - Vernier (L.C.: 0.02 mm)	Using Caliper Checker by Comparison Method	0 to 600 mm	15.01 μ m
135	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Caliper - Vernier / Dial (L.C.: 0.02 mm)	Using Caliper Checker by Comparison Method	0 to 150 mm	15 μ m



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136	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Caliper - Vernier / Dial (L.C.: 0.02 mm)	Using Caliper Checker by Comparison Method	0 to 300 mm	15 µm
137	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Caliper - Vernier / Dial (L.C.: 0.02 mm)	Using Caliper Checker by Comparison Method	0 to 600 mm	15 µm
138	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Caliper - Vernier / Dial (L.C.: 0.05 mm)	Using Caliper Checker by Comparison Method	0 to 300 mm	30 µm
139	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Caliper - Vernier / Dial (L.C.: 0.05 mm)	Using Caliper Checker by Comparison Method	0 to 600 mm	30 µm
140	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Caliper - Vernier / Dial (L.C.: 0.05 mm)	Using Caliper Checker by Comparison Method	0 to 150 mm	30 µm
141	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Caliper - Vernier / Dial / Digital (L.C.: 0.01 mm)	Using Caliper Checker by Comparison Method	0 to 300 mm	13 µm



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142	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Caliper - Vernier / Dial / Digital (L.C.: 0.01 mm)	Using Caliper Checker by Comparison Method	0 to 150 mm	9.3 µm
143	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Coating Thickness Gauge (L.C.: 0.1 µm)	Using Standard Foils by Comparison Method	25 µm to 745 µm	4 µm
144	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Core Cutter - Linearity	Using Digital Vernier Caliper by Comparison Method	Up to 150 mm	15 µm
145	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Cube Mould - Linearity	Using Digital Vernier Caliper by Comparison Method	Up to 150 mm	15 µm
146	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Cylindrical Measuring Pin	Using Slip Gauge Set, Plunger Dial Gauge with Comparator Stand by Comparison Method	0.5 mm to 12 mm	3.16 µm
147	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Depth Micrometer (L.C.: 0.01 mm)	Using Slip Gauge Set by Comparison Method	0 to 25 mm	9.36 µm



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148	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Dial Thickness Gauge (L.C.: 0.01 mm)	Using Slip Gauge Set by Comparison Method	0 to 10 mm	7.2 µm
149	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Dial Thickness Gauge (L.C.: 0.001 mm)	Using Slip Gauge Set by Comparison Method	0 to 1 mm	1 µm
150	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Dial Thickness Gauge (L.C.: 0.01 mm)	Using Slip Gauge Set by Comparison Method	0 to 20 mm	9.2 µm
151	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Elongation Gauge - Linear	Using Vernier Caliper by Comparison Method	14.7 mm to 81 mm	15 µm
152	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	External Micrometer (L.C.: 0.001 mm)	Using Slip Gauge Set by Comparison Method	0 to 100 mm	3.8 µm
153	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	External Micrometer (L.C.: 0.001 mm)	Using Slip Gauge Set by Comparison Method	0 to 25 mm	1.8 µm



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154	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	External Micrometer (L.C.: 0.01 mm)	Using Slip Gauge Set by Comparison Method	0 to 100 mm	7 µm
155	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	External Micrometer (L.C.: 0.01 mm)	Using Slip Gauge Set by Comparison Method	0 to 25 mm	6 µm
156	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Feeler Gauge	Using Digital Micrometer by Comparison Method	0.05 mm to 1 mm	1.76 µm
157	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Flakiness Gauge - Linear	Using Digital Vernier Caliper by Comparison Method	4.89 mm to 33.9 mm	12.8 µm
158	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Height Gauge - Dial / Vernier (L.C.: 0.02 mm)	Using Caliper Checker, Surface Plate by Comparison Method	0 to 600 mm	13 µm
159	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Height Gauge - Dial / Vernier (L.C.: 0.02 mm)	Using Caliper Checker, Surface Plate by Comparison Method	0 to 300 mm	15.6 µm



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160	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Height Gauge - Vernier / Dial / Digital (L.C.: 0.01 mm)	Using Caliper Checker, Surface Plate by Comparison Method	0 to 300 mm	9.9 µm
161	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Height Gauge - Vernier / Dial / Digital (L.C.: 0.01 mm)	Using Caliper Checker, Surface Plate by Comparison Method	0 to 600 mm	13.4 µm
162	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Internal Micrometer (L.C.: 0.01 mm)	Using Slip Gauge Set, Slip Gauge Accessories by Comparison Method	50 mm to 100 mm	8 µm
163	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Metric Steel / Woven Metallic / Fiber Tape (L.C.: 1 mm)	Using Scale & Tape Calibration Unit by Comparison Method	0 to 10000 mm	207.3 x sqrt (L) µm, Where L is in m
164	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Metric Steel Scale (L.C.: 1 mm)	Using Scale & Tape Calibration Unit by Comparison Method	0 to 1000 mm	133 x Sqrt (L) µm, where L is in m
165	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Plain Plug Gauge	Using Slip Gauge Set, Plunger Dial Gauge with Comparator Stand by Comparison Method	3 mm to 100 mm	13.2 µm



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166	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Plunger Type Dial Gauge - Dial / Digital (L.C.: 0.001 mm)	Using Slip Gauge Set with Comparator Stand by Direct Method	0 to 25 mm	2.44 µm
167	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Plunger Type Dial Gauge - Dial / Digital (L.C.: 0.001 mm)	Using Slip Gauge Set with Comparator Stand by Direct Method	0 to 10 mm	2.5 µm
168	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Plunger Type Dial Gauge - Dial / Digital (L.C.: 0.01 mm)	Using Slip Gauge Set with Comparator Stand by Direct Method	0 to 50 mm	7 µm
169	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Radius Gauge	Using Profile Projector by Comparison Method	7.5 mm to 15 mm	6.7 µm
170	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Setting Rod	Using Slip Gauge Set, Plunger Dial Gauge with Comparator Stand Comparison Method	25 mm to 100 mm	9 µm
171	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Slump Cone - Linearity	Using Digital Vernier Caliper by Comparison Method	Up to 300 mm	18 µm



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172	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Snap Gauge	Using Slip Gauge Set by Comparison Method	1 mm to 100 mm	7.58 µm
173	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Taper Scale (L.C.: 0.1 mm)	Using Profile Projector by Comparison Method	1 mm to 15 mm	8.78 µm
174	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Test Sieve - Aperture Size	Using Digital Caliper by Comparison Method	4 mm to 125 mm	24 µm
175	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Thread Pitch Gauge - Flank Angle	Using Profile Projector by Comparison Method	Up to 60 °	0.2 minute of arc
176	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Thread Pitch Gauge - Linear	Using Profile Projector by Comparison Method	0.25 mm to 6 mm	7.8 µm
177	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Ultrasonic Thickness Gauge (L.C.: 0.1 mm)	Using Gauge Block Set by Comparison Method	1.2 mm to 100 mm	78.5 µm



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178	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Weld Gauge - Angle	Using Profile Projector by Comparison Method	0 ° to 90 °	6 minute of arc
179	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Weld Gauge - Linear	Using Profile Projector by Comparison Method	0 to 60 mm	10 µm
180	MECHANICAL-DIMENSION (BASIC MEASURING INSTRUMENT, GAUGE ETC.)	Wire Gauge	Using Profile Projector by Comparison Method	0.19 mm to 7.62 mm	9 µm
181	MECHANICAL-MOBILE FORCE MEASURING SYSTEM	Push Pull Gauge, Force Gauge - Push & Pull Mode	Using Newton Weights by Comparison Method as per VDI/VDE 2624 Part 2.1 : 2008	10 N to 100 N	3.3 %
182	MECHANICAL-MOBILE FORCE MEASURING SYSTEM	Push Pull Gauge, Force Gauge - Push & Pull Mode	Using Newton Weights by Comparison Method as per VDI/VDE 2624 Part 2.1 : 2008	> 100 N to 1000 N	1.8 %
183	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Manometer, Differential Pressure Gauge, Magnehelic Gauge, Pressure Transmitter - Pneumatic Medium	Using Digital Manometer, Portable Process Calibrator, Pneumatic Pump, Digital Multimeter by Comparison Method as per DKD-R 6-1	0 to 950 mbar	0.83 mbar



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184	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Pressure Gauge, Pressure Switch, Pressure Transmitter with Indicator - Hydraulic Pressure	Using Digital Pressure Gauge, Portable Process Calibrator, Hydraulic Comparator, Digital Multimeter by Comparison Method as per DKD-R 6-1	0 to 700 bar	0.41 bar
185	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Pressure Gauge, Pressure Switch, Pressure Transmitter with Indicator - Pneumatic Pressure	Using Digital Pressure Gauge, Portable Process Calibrator, Pneumatic Pump, Digital Multimeter by Comparison Method as per DKD-R 6-1	0 to 2 bar	0.002 bar
186	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Pressure Gauge, Pressure Switch, Pressure Transmitter with Indicator - Pneumatic Pressure	Using Digital Pressure Gauge, Portable Process Calibrator, Pneumatic Comparator, Digital Multimeter by Comparison Method as per DKD-R 6-1	0 to 30 bar	0.1 bar
187	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Vacuum Gauge, Vacuum Switch, Vacuum Transmitter, Vacuum Transducer - Pneumatic Pressure	Using Digital Pressure Gauge with Pneumatic Pump and Digital Multimeter by Comparison Method as per DKD-R 6-1	(-) 0.9 bar to 0 bar	0.006 bar
188	MECHANICAL-UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Compression Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	1000 N to 5000 N	0.67 %



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189	MECHANICAL- UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Compression Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	5000 N to 100 kN	0.6 %
190	MECHANICAL- UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Compression Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	20 N to 1000 N	0.67 %
191	MECHANICAL- UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Tension Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	1000 N to 5000 N	0.67 %
192	MECHANICAL- UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Tension Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	5000 N to 100 kN	0.6 %
193	MECHANICAL- UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Tension Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	20 N to 1000 N	0.67 %



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194	MECHANICAL-VOLUME	Glass Burette	Using Digital Precision Balance (Readability : 0.1 mg) and Distilled Water of Known Density by Gravimetric Method as per ISO 4787:2021 & ISO/TR 20461	1 ml to 100 ml	0.2 ml
195	MECHANICAL-VOLUME	Glass Pipette (Graduated / Non Graduated)	Using Digital Precision Balance (Readability : 0.1 mg) and Distilled Water of Known Density by Gravimetric Method as per ISO 4787:2021 & ISO/TR 20461	> 25 ml to 100 ml	0.05 ml
196	MECHANICAL-VOLUME	Glass Pipette (Graduated / Non Graduated)	Using Digital Precision Balance (Readability : 0.1 mg) and Distilled Water of Known Density by Gravimetric Method as per ISO 4787:2021 & ISO/TR 20461	1 ml to 25 ml	0.048 ml
197	MECHANICAL-VOLUME	Measuring Cylinder, Volumetric Flask, Beaker, Jar, Conical Flask	Using Weighing Balance (Readability : 0.1 mg) and Distilled Water of known Density by Gravimetric Method as per ISO 4787:2021	1 ml to 100 ml	0.24 ml



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198	MECHANICAL-VOLUME	Micropipette	Using Analytical Balance (Readability : 0.1 mg) & Distilled Water by Gravimetric Method per ISO 8655-6:2022	200 µl to 1000 µl	11.42 µl
199	MECHANICAL-VOLUME	Pipette, Burette, Measuring Cylinder, Volumetric Flask, Beaker, Jar, Conical Flask	Using Weighing Balance (Readability : 1 g) and Distilled Water of known Density by Gravimetric Method as per ISO 4787:2021	> 5000 ml to 8000 ml	9.45 ml
200	MECHANICAL-VOLUME	Pipette, Burette, Measuring Cylinder, Volumetric Flask, Beaker, Jar, Conical Flask	Using Weighing Balance of (Readability : 1 mg) and Distilled Water of known Density by Gravimetric Method as per ISO 4787:2021	> 100 ml to 1000 ml	0.4 ml
201	MECHANICAL-VOLUME	Pipette, Burette, Measuring Cylinder, Volumetric Flask, Beaker, Jar, Conical Flask	Using Weighing Balance of (Readability : 1 mg) and Distilled Water of known Density by Gravimetric Method as per ISO 4787:2021	> 1000 ml to 2000 ml	1.71 ml
202	MECHANICAL-VOLUME	Pipette, Burette, Measuring Cylinder, Volumetric Flask, Beaker, Jar, Conical Flask	Using Weighing Balance of (Readability : 10 mg) and Distilled Water of known Density by Gravimetric Method as per ISO 4787:2021	> 2000 to 5000 ml	1.71 ml



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203	MECHANICAL-WEIGHING SCALE AND BALANCE	Electronic Balance - Accuracy Class I and Coarser (Readability : 0.1 mg)	Using E1 Class Weights by Comparison Method as per OIML R 76-1	0 to 220 g	0.3 mg
204	MECHANICAL-WEIGHING SCALE AND BALANCE	Electronic Balance - Accuracy Class III and Coarser (Readability : 1 g)	Using F1 Class Weights by Comparison Method as per OIML R 76-1	0 to 35 kg	1.6 g
205	MECHANICAL-WEIGHING SCALE AND BALANCE	Electronic Balance - Accuracy Class III and Coarser (Readability : 1 g)	Using F1 Class Weights by Comparison Method as per OIML R 76-1	0 to 100 kg	7.23 g
206	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	1 g	0.1 mg
207	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using F1 Class Weight and Weighing Balance (Readability : 0.001 g) by Substitution Method (ABBA Cycle) as per OIML R 111-1	1 kg	2 mg
208	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	10 g	0.18 mg



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209	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	100 g	0.24 mg
210	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	2 g	0.13 mg
211	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using F1 Class Weight and Weighing Balance (Readability : 0.001 g) by Substitution Method (ABBA Cycle) as per OIML R 111-1	2 kg	8 mg
212	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	20 g	0.21 mg



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213	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	200 g	0.36 mg
214	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	5 g	0.15 mg
215	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using F1 Class Weight and Weighing Balance (Readability : 0.01 g) by Substitution Method (ABBA Cycle) as per OIML R 111-1	5 kg	20 mg
216	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	50 g	0.24 mg



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217	MECHANICAL-WEIGHTS	Accuracy Class F2 & Coarser	Using F1 Class Weight and Weighing Balance (Readability : 0.001 g) by Substitution Method (ABBA Cycle) as per OIML R 111-1	500 g	2 mg
218	MECHANICAL-WEIGHTS	Accuracy Class M1 & Coarser	Using F1 Class Weight and Weighing Balance (Readability : 0.01 g) by Substitution Method (ABBA Cycle) as per OIML R 111-1	10 kg	123 mg
219	MECHANICAL-WEIGHTS	Accuracy Class M1 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	10 mg	0.08 mg
220	MECHANICAL-WEIGHTS	Accuracy Class M1 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	100 mg	0.11 mg



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221	MECHANICAL-WEIGHTS	Accuracy Class M1 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	20 mg	0.1 mg
222	MECHANICAL-WEIGHTS	Accuracy Class M1 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	200 mg	0.12 mg
223	MECHANICAL-WEIGHTS	Accuracy Class M1 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	50 mg	0.11 mg
224	MECHANICAL-WEIGHTS	Accuracy Class M1 & Coarser	Using E1 Class Weight and Weighing Balance (Readability : 0.1 mg) by Substitution Method (ABBA Cycle) as per OIML R 111-1	500 mg	0.12 mg



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225	MECHANICAL-WEIGHTS	Accuracy Class M1-2 & Coarser	Using F1 Class Weight and Weighing Balance (Readability : 1 g) by Substitution Method (ABBA Cycle) as per OIML R 111-1	100 kg	2223 mg
226	MECHANICAL-WEIGHTS	Accuracy Class M1-2 & Coarser	Using F1 Class Weight and Weighing Balance (Readability : 1 g) by Substitution Method (ABBA Cycle) as per OIML R 111-1	50 kg	1668 mg
227	MECHANICAL-WEIGHTS	Accuracy Class M2 & Coarser	Using F1 Class Weight and Weighing Balance (Readability : 1 g) by Substitution Method (ABBA Cycle) as per OIML R 111-1	20 kg	720 mg
228	THERMAL-SPECIFIC HEAT & HUMIDITY	Digital /Analog Hygrometer, Temperature and RH Sensor with Indicator / Controller / Data Logger/ Recorder @ 25 °C	Using Digital Thermo Hygrometer With Sensor & Humidity Chamber by Comparison Method	10 % RH to 95 % RH	1.2 % RH
229	THERMAL-SPECIFIC HEAT & HUMIDITY	Digital /Analog Hygrometer, Temperature and RH Sensor with Indicator / Controller / Data Logger/ Recorder @ 50 % RH	Using Digital Thermo Hygrometer With Sensor & Humidity Chamber by Comparison Method	10 °C to 50 °C	0.71 °C



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230	THERMAL-SPECIFIC HEAT & HUMIDITY	Temperature / Humidity Indicator with Sensor of Humidity / Environmental Chamber - Single Position @ 25°C	Using Digital Thermo Hygrometer with Sensor by Comparison Method	10 % RH to 90 % RH	1.8 % RH
231	THERMAL-TEMPERATURE	Liquid in Glass Thermometer	Using 4 Wire RTD Sensor with Indicator, 6½ Digit Multimeter, Liquid Bath by Comparison Method	(-) 30 °C to 50 °C	0.65 °C
232	THERMAL-TEMPERATURE	Dry Block Furnace, Muffle Furnace - Multi Position (Minimum 9 Sensors)	Using N - Type Thermocouple with Data Logger by Comparison Method	300 °C to 600 °C	3.2 °C
233	THERMAL-TEMPERATURE	Environment Chamber, Furnace, Freezer, Oven, Vacuum Oven, Cold Chamber, Liquid Bath, Cold / Hot Room, Aging Oven - Multi Position (Minimum 9 Sensors)	Using RTD Sensor with Data Logger by Comparison Method	0 °C to 300 °C	1.5 °C
234	THERMAL-TEMPERATURE	Environment Chamber, Furnace, Freezer, Walking Chiller, Cold Chamber, Liquid Bath, Cold Room - Multi Position (Minimum 9 Sensors)	Using RTD Sensor with Data Logger by Comparison Method	(-) 80 °C to 0 °C	2.13 °C



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235	THERMAL-TEMPERATURE	Liquid in Glass Thermometer	Using 4 Wire RTD Sensor with Indicator, 6½ Digit Multimeter, Liquid Bath by Comparison Method	50 °C to 250 °C	0.65 °C
236	THERMAL-TEMPERATURE	RTD / Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Temperature Transducer, Temperature Gauge, Digital Thermometer	Using 4 Wire RTD Sensor with Indicator, 6½ Digital Multimeter, Liquid Bath by Comparison Method	> 50 °C to 250 °C	0.57 °C
237	THERMAL-TEMPERATURE	RTD / Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Temperature Transducer, Temperature Gauge, Digital Thermometer	Using 4 Wire RTD Sensor with Indicator, 6½ Digital Multimeter, Liquid Bath by Comparison Method	(-) 30 °C to 50 °C	0.52 °C
238	THERMAL-TEMPERATURE	Temperature Indicator / Controller with Sensor of Liquid Bath, Oven, Dry Block, Furnace, Melting Point Apparatus, Environmental Chamber - Single Position	Using S - Type Thermocouple with Indicator, 6½ Digit Multimeter by Comparison Method	300 °C to 600 °C	1.08 °C



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239	THERMAL-TEMPERATURE	Temperature Indicator with Sensor of Dry Block Furnace, Muffle Furnace - Single Position	Using S - Type Thermocouple with Indicator, 6½ Digit Multimeter by Comparison Method	> 800 °C to 1200 °C	2.6 °C
240	THERMAL-TEMPERATURE	Temperature Indicator with Sensor of Dry Block Furnace, Muffle Furnace - Single Position	Using S - Type Thermocouple with Indicator, 6½ Digit Multimeter by Comparison Method	600 °C to 800 °C	1.9 °C
241	THERMAL-TEMPERATURE	Temperature Indicator with Sensor of Liquid Bath, Oven, Dry Block Furnace, Refrigerator, Autoclave, Incubator (Non Medical Application), Environmental Chamber, Salt Spray Chamber - Single Position	Using 4 Wire RTD with Indicator by Comparison Method	(-) 50 °C to 300 °C	0.67 °C
242	THERMAL-TEMPERATURE	Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Digital Thermometer	Using S - Type Thermocouple with Indicator, 6½ Digit Multimeter by Comparison Method	600 °C to 800 °C	1.8 °C



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243	THERMAL-TEMPERATURE	Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Digital Thermometer	Using S - Type Thermocouple with Indicator, 6½ Digital Multimeter by Comparison Method	> 800 °C to 1200 °C	2.6 °C
244	THERMAL-TEMPERATURE	Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Temperature Transducer, Temperature Gauge ,Digital Thermometer	Using S - Type Thermocouple with Indicator, 6½ Digital Multimeter by Comparison Method	> 250 °C to 600 °C	1.19 °C



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Site Facility					
1	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Current @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 A to 10 A	0.39 % to 0.18 %
2	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Current @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 mA to 100 mA	0.76 % to 0.57 %
3	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Current @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	100 mA to 1 A	0.57 % to 0.39 %
4	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Resistance @ 1 kHz - 2 Wire	Using LCR Meter by Direct Method	1 ohm to 100 kohm	0.15 % to 0.08 %
5	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Voltage @ 50 Hz	Using Digital Multimeter with High Voltage Probe by Direct Method	1 kV to 10 kV	3 %
6	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Voltage @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 V to 750 V	0.86 %



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7	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Voltage @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	10 mV to 100 mV	0.55 % to 0.63 %
8	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	AC Voltage @ 50 Hz	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	100 mV to 1 V	0.63 % to 0.86 %
9	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	Capacitance @ 1 kHz	Using LCR Meter by Direct Method	100 pF to 100 µF	0.55 % to 0.56 %
10	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Measure)	Inductance @ 1 kHz	Using LCR Meter by Direct Method	100 µH to 10 H	0.68 % to 0.82 %
11	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Source)	1Ø, 4 Wire, AC Active Energy @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.005 kWh to 0.9 kWh	0.25 %
12	ELECTRO-TECHNICAL-Alternating Current (< 1 GHz) (Source)	1Ø, 4 Wire, AC Active Power @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.01 kW to 1.8 kW	0.27 %



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13	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	1Ø, 4 Wire, AC Reactive Energy @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.008 kVarh to 0.783 kVarh	0.25 %
14	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	1Ø, 4 Wire, AC Reactive Power @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.017 kVAr to 1.558 kVAr	0.27 %
15	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	3Ø, 4 Wire, AC Active Energy @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.015 kWh to 2.7 kWh	0.25 %
16	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	3Ø, 4 Wire, AC Active Power @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.03 kW to 5.4 kW	0.27 %
17	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	3Ø, 4 Wire, AC Reactive Energy @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.024 kVarh to 2.349 kVarh	0.25 %
18	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	3Ø, 4 Wire, AC Reactive Power @ (50 Hz, 0.5 Lead / Lag to UPF, 40 V to 300 V, 0.5 A to 6 A)	Using Standard 3Ø Power Energy Calibrator by Comparison Method	0.052 kVAr to 4.677 kVAr	0.27 %
19	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator by Direct Method	1 A to 10 A	0.5 % to 0.48 %



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20	ELECTRO- TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator by Direct Method	1 mA to 100 mA	1 % to 0.4 %
21	ELECTRO- TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator with 100 Turns Current Coil by Direct Method	10 A to 100 A	0.48 % to 2 %
22	ELECTRO- TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator by Direct Method	100 mA to 1 A	0.4 % to 0.5 %
23	ELECTRO- TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator by Direct Method	100 µA to 1 mA	0.79 % to 1 %
24	ELECTRO- TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Current @ 50 Hz	Using Multi Function Calibrator with Current Coil by Direct Method	100 A to 1000 A	2 % to 0.8 %
25	ELECTRO- TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Voltage @ 50 Hz	Using Multi Function Calibrator by Direct Method	100 V to 1000 V	0.27 % to 0.36 %
26	ELECTRO- TECHNICAL- Alternating Current (< 1 GHz) (Source)	AC Voltage @ 50 Hz	Using Multi Function Calibrator by Direct Method	20 mV to 100 V	1.08 % to 0.27 %



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27	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Capacitance @ 1 kHz	Using Capacitance Box by Direct Method	1 nF to 10 μ F	0.46 % to 0.16 %
28	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Inductance @ 1 kHz	Using Inductance Box by Direct Method	1 mH to 10 mH	0.73 % to 0.84 %
29	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Inductance @ 1 kHz	Using Inductance Box by Direct Method	10 mH to 1 H	0.84 % to 0.61 %
30	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Power Factor @ 50 Hz	Using Standard 3 $\emptyset$ Energy Source by Direct Method	0.1 (Lead / Lag) to UPF	0.008 PF
31	ELECTRO-TECHNICAL- Alternating Current (< 1 GHz) (Source)	Resistance @ 1 kHz - 2 Wire	Using Decade Resistance Box by Direct Method	1 ohm to 100 kohm	0.5 % to 0.58 %
32	ELECTRO-TECHNICAL- DIRECT CURRENT (Measure)	Resistance - 4 Wire	Using Digital Micro Ohm Meter by Direct Method	20 μ ohm to 20 kohm	0.9 % to 0.65 %
33	ELECTRO-TECHNICAL- DIRECT CURRENT (Measure)	DC Current	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 A to 10 A	0.12 % to 0.19 %



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34	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Current	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 mA to 1 A	0.76 % to 0.12 %
35	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC High Voltage	Using Digital Multimeter with High Voltage Probe by Direct Method	1 kV to 10 kV	2.4 %
36	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	1 kohm to 10 kohm	0.37 % to 0.13 %
37	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	1 Mohm to 10 Mohm	0.13 % to 0.05 %
38	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	10 kohm to 1 Mohm	0.13 %
39	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	10 Mohm to 100 Mohm	0.05 % to 1.51 %
40	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	10 ohm to 100 ohm	0.82 % to 0.13 %



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41	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Resistance - 2 Wire	Using 6½ Digit Multimeter by Direct Method	100 ohm to 1 kohm	0.13 % to 0.37 %
42	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Voltage	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	1 V to 100 V	0.18 % to 0.11 %
43	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Voltage	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	10 mV to 100 mV	0.71 % to 0.18 %
44	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Voltage	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	100 mV to 1 V	0.18 %
45	ELECTRO-TECHNICAL-DIRECT CURRENT (Measure)	DC Voltage	Using 6½ Digit Multimeter, Multi Function Calibrator by Comparison Method	100 V to 1000 V	0.11 % to 0.1 %
46	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator by Direct Method	1 A to 10 A	0.5 %
47	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator by Direct Method	1 mA to 100 mA	0.3 %



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48	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator with 100 Turn Current Coil by Direct Method	10 A to 100 A	0.5 % to 2 %
49	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator with 100 Turn Current Coil by Direct Method	100 A to 1000 A	2 % to 0.95 %
50	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Current	Using Multi Function Calibrator by Direct Method	100 mA to 1 A	0.3 % to 0.5 %
51	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Voltage	Using Multi Function Calibrator by Direct Method	10 mV to 100 mV	0.31 %
52	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Voltage	Using Multi Function Calibrator by Direct Method	10 V to 100 V	0.21 % to 0.19 %
53	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Voltage	Using Multi Function Calibrator by Direct Method	100 mV to 10 V	0.31 % to 0.21 %
54	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	DC Voltage	Using Multi Function Calibrator by Direct Method	100 V to 1000 V	0.19 %



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55	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box (Discrete Values) by Direct Method	0.001 ohm to 1 kohm	1.61 %
56	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box by Direct Method	1 kohm to 5 kohm	1.61 %
57	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Mega Ohm Box by Direct Method	1 Mohm to 5 Mohm	0.8 %
58	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box (Discrete Values) by Direct Method	100 kohm to 1000 kohm	0.13 % to 0.8 %
59	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box by Direct Method	5 kohm to 10 kohm	1.61 % to 0.12 %
60	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box (Discrete Values) by Direct Method	5 Mohm to 100 Mohm	0.8 % to 1.5 %
61	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box by Direct Method	50 kohm to 100 kohm	0.12 % to 0.13 %



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62	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using Resistance Box & Decade Resistance Box (Discrete Values) by Direct Method	100 Mohm to 20 Gohm	1.5 % to 2.75 %
63	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire	Using Resistance Box & Decade Resistance Box by Direct Method	10 kohm to 50 kohm	0.12 %
64	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using HV Mega Ohm Box (Discrete Values) by Direct Method	2 Gohm to 20 Gohm	2.25 % to 2.75 %
65	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using HV Mega Ohm Box (Discrete Values) by Direct Method	20 Mohm to 200 Mohm	1.5 % to 1.75 %
66	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using HV Mega Ohm Box (Discrete Values) by Direct Method	200 Mohm to 2 Gohm	1.75 % to 2.25 %
67	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 2 Wire @ 1000 V	Using HV Mega Ohm Box (Discrete Values) by Direct Method	2 Mohm to 20 Mohm	0.8 % to 1.5 %
68	ELECTRO-TECHNICAL-DIRECT CURRENT (Source)	Resistance - 4 Wire	Using Standard Resistance Box (Discrete Values) by Direct Method	20 µohm to 100 kohm	0.47 % to 0.12 %



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69	ELECTRO-TECHNICAL-ELECTRICAL EQUIPMENT (Measure)	AC Energy 3Ø, 4 Wire @ (50 Hz, UPF to 0.5 lag, 240 V, 1.25 A to 5 A)	Using Standard Energy Meter by Direct Method	0.45 kWh to 3.6 kWh	0.38 %
70	ELECTRO-TECHNICAL-ELECTRICAL EQUIPMENT (Measure)	AC Power, 3Ø, 4 Wire @ (50 Hz, UPF to 0.5 lag, 240 V, 1.25 A to 5 A)	Using Standard Energy Meter by Direct Method	0.45 kW to 3.6 kW	0.36 %
71	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	RTD (PT 100)	Using Universal Calibrator by Direct Method	(-) 100 °C to 650 °C	0.83 °C
72	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	Thermocouple B Type	Using Universal Calibrator by Direct Method	0 °C to 600 °C	0.82 °C
73	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	Thermocouple E Type	Using Universal Calibrator by Direct Method	(-) 100 °C to 1000 °C	0.91 °C
74	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	Thermocouple J Type	Using Universal Calibrator by Direct Method	(-) 100 °C to 700 °C	0.83 °C
75	ELECTRO-TECHNICAL-TEMPERATURE SIMULATION (Measure)	Thermocouple K Type	Using Universal Calibrator by Direct Method	(-) 200 °C to 1300 °C	0.83 °C



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76	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Measure)	Thermocouple T Type	Using Universal Calibrator by Direct Method	0 °C to 400 °C	0.92 °C
77	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	RTD (PT 100)	Using Universal Calibrator by Direct Method	(-) 100 °C to 650 °C	0.96 °C
78	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple B Type	Using Universal Calibrator by Direct Method	0 °C to 600 °C	0.59 °C
79	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple E Type	Using Universal Calibrator by Direct Method	(-) 100 °C to 1000 °C	0.83 °C
80	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple J Type	Using Universal Calibrator by Direct Method	(-) 200 °C to 700 °C	0.83 °C
81	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple K Type	Using Universal Calibrator by Direct Method	(-) 200 °C to 1300 °C	0.84 °C
82	ELECTRO- TECHNICAL- TEMPERATURE SIMULATION (Source)	Thermocouple T Type	Using Universal Calibrator by Direct Method	0 °C to 400 °C	0.83 °C



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83	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Frequency	Using 6½ Digit Multimeter by Direct Method	10 Hz to 100 Hz	0.58 % to 0.06 %
84	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Frequency	Using 6½ Digit Multimeter by Direct Method	100 Hz to 100 kHz	0.06 % to 0.03 %
85	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Frequency	Using 6½ Digit Multimeter by Direct Method	100 kHz to 1000 kHz	0.03 %
86	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Time	Using Time Interval Meter by Comparison Method	1 hour to 8 hour	0.44 s to 11.8 s
87	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Time	Using Time Interval Meter by Comparison Method	1 minute to 1 hour	0.017 s to 0.44 s
88	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Time	Using Time Interval Meter by Comparison Method	10 s to 1 minute	1.9 ms to 0.017 s
89	ELECTRO- TECHNICAL- TIME & FREQUENCY (Measure)	Time	Using Time Interval Meter by Comparison Method	8 hour to 24 hour	11.8 s to 23 s



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90	FLUID FLOW-FLOW MEASURING DEVICES	Dial Flow Meter, Digital Flow Meter, Flow Transmitter, Flow Transducer (Medium : Water)	Using Ultrasonic Flow Meter by Comparison Method	> 100 m ³ /h to 250 m ³ /h	1.09 %
91	FLUID FLOW-FLOW MEASURING DEVICES	Dial Flow Meter, Digital Flow Meter, Flow Transmitter, Flow Transducer (Medium : Water)	Using Ultrasonic Flow Meter by Comparison Method	1 m ³ /h to 100 m ³ /h	1.09 %
92	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Manometer, Differential Pressure Gauge, Magnehelic Gauge, Pressure Transmitter - Pneumatic Medium	Using Digital Manometer, Portable Process Calibrator, Pneumatic Pump, Digital Multimeter by Comparison Method as per DKD-R 6-1	0 to 950 mbar	0.83 mbar
93	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Pressure Gauge, Pressure Switch, Pressure Transmitter with Indicator - Hydraulic Pressure	Using Digital Pressure Gauge, Portable Process Calibrator, Hydraulic Comparator, Digital Multimeter by Comparison Method as per DKD-R 6-1	0 to 700 bar	0.41 bar
94	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Pressure Gauge, Pressure Switch, Pressure Transmitter with Indicator - Pneumatic Pressure	Using Digital Pressure Gauge, Portable Process Calibrator, Pneumatic Pump, Digital Multimeter by Comparison Method as per DKD-R 6-1	0 to 2 bar	0.002 bar



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95	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Pressure Gauge, Pressure Switch, Pressure Transmitter with Indicator - Pneumatic Pressure	Using Digital Pressure Gauge, Portable Process Calibrator, Pneumatic Comparator, Digital Multimeter by Comparison Method as per DKD-R 6-1	0 to 30 bar	0.1 bar
96	MECHANICAL-PRESSURE INDICATING DEVICES	Analog / Digital - Vacuum Gauge, Vacuum Switch, Vacuum Transmitter, Vacuum Transducer - Pneumatic Pressure	Using Digital Pressure Gauge with Pneumatic Pump and Digital Multimeter by Comparison Method as per DKD-R 6-1	(-) 0.9 bar to 0 bar	0.006 bar
97	MECHANICAL-UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Compression Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	1000 N to 5000 N	0.67 %
98	MECHANICAL-UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Compression Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	5000 N to 100 kN	0.6 %
99	MECHANICAL-UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Compression Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	20 N to 1000 N	0.67 %



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100	MECHANICAL-UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Tension Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	1000 N to 5000 N	0.67 %
101	MECHANICAL-UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Tension Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	5000 N to 100 kN	0.6 %
102	MECHANICAL-UTM, TENSION CREEP AND TORSION TESTING MACHINE	Universal Testing Machine - Tension Mode (Class I and Coarser)	Using Force Proving Instruments (Load Cell with Indicator) by Comparison Method as per IS 1828 (Part 1) : 2022	20 N to 1000 N	0.67 %
103	THERMAL-SPECIFIC HEAT & HUMIDITY	Digital /Analog Hygrometer, Temperature and RH Sensor with Indicator / Controller / Data Logger/ Recorder @ 25 °C	Using Digital Thermo Hygrometer With Sensor & Humidity Chamber by Comparison Method	10 % RH to 95 % RH	1.2 % RH
104	THERMAL-SPECIFIC HEAT & HUMIDITY	Digital /Analog Hygrometer, Temperature and RH Sensor with Indicator / Controller / Data Logger/ Recorder @ 50 % RH	Using Digital Thermo Hygrometer With Sensor & Humidity Chamber by Comparison Method	10 °C to 50 °C	0.71 °C



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105	THERMAL-SPECIFIC HEAT & HUMIDITY	Temperature / Humidity Indicator with Sensor of Humidity / Environmental Chamber - Single Position @ 25°C	Using Digital Thermo Hygrometer with Sensor by Comparison Method	10 % RH to 90 % RH	1.8 % RH
106	THERMAL-TEMPERATURE	Liquid in Glass Thermometer	Using 4 Wire RTD Sensor with Indicator, 6½ Digit Multimeter, Liquid Bath by Comparison Method	(-) 30 °C to 50 °C	0.65 °C
107	THERMAL-TEMPERATURE	Dry Block Furnace, Muffle Furnace - Multi Position (Minimum 9 Sensors)	Using N - Type Thermocouple with Data Logger by Comparison Method	300 °C to 600 °C	3.2 °C
108	THERMAL-TEMPERATURE	Environment Chamber, Furnace, Freezer, Oven, Vacuum Oven, Cold Chamber, Liquid Bath, Cold / Hot Room, Aging Oven - Multi Position (Minimum 9 Sensors)	Using RTD Sensor with Data Logger by Comparison Method	0 °C to 300 °C	1.5 °C
109	THERMAL-TEMPERATURE	Environment Chamber, Furnace, Freezer, Walking Chiller, Cold Chamber, Liquid Bath, Cold Room - Multi Position (Minimum 9 Sensors)	Using RTD Sensor with Data Logger by Comparison Method	(-) 80 °C to 0 °C	2.13 °C



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110	THERMAL-TEMPERATURE	Liquid in Glass Thermometer	Using 4 Wire RTD Sensor with Indicator, 6½ Digit Multimeter, Liquid Bath by Comparison Method	50 °C to 250 °C	0.65 °C
111	THERMAL-TEMPERATURE	RTD / Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Temperature Transducer, Temperature Gauge, Digital Thermometer	Using 4 Wire RTD Sensor with Indicator, 6½ Digital Multimeter, Liquid Bath by Comparison Method	> 50 °C to 250 °C	0.57 °C
112	THERMAL-TEMPERATURE	RTD / Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Temperature Transducer, Temperature Gauge, Digital Thermometer	Using 4 Wire RTD Sensor with Indicator, 6½ Digital Multimeter, Liquid Bath by Comparison Method	(-) 30 °C to 50 °C	0.52 °C
113	THERMAL-TEMPERATURE	Temperature Indicator / Controller with Sensor of Liquid Bath, Oven, Dry Block, Furnace, Melting Point Apparatus, Environmental Chamber - Single Position	Using S - Type Thermocouple with Indicator, 6½ Digit Multimeter by Comparison Method	300 °C to 600 °C	1.08 °C



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114	THERMAL-TEMPERATURE	Temperature Indicator with Sensor of Dry Block Furnace, Muffle Furnace - Single Position	Using S - Type Thermocouple with Indicator, 6½ Digit Multimeter by Comparison Method	> 800 °C to 1200 °C	2.6 °C
115	THERMAL-TEMPERATURE	Temperature Indicator with Sensor of Dry Block Furnace, Muffle Furnace - Single Position	Using S - Type Thermocouple with Indicator, 6½ Digit Multimeter by Comparison Method	600 °C to 800 °C	1.9 °C
116	THERMAL-TEMPERATURE	Temperature Indicator with Sensor of Liquid Bath, Oven, Dry Block Furnace, Refrigerator, Autoclave, Incubator (Non Medical Application), Environmental Chamber, Salt Spray Chamber - Single Position	Using 4 Wire RTD with Indicator by Comparison Method	(-) 50 °C to 300 °C	0.67 °C
117	THERMAL-TEMPERATURE	Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Digital Thermometer	Using S - Type Thermocouple with Indicator, 6½ Digit Multimeter by Comparison Method	600 °C to 800 °C	1.8 °C



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118	THERMAL-TEMPERATURE	Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Digital Thermometer	Using S - Type Thermocouple with Indicator, 6½ Digital Multimeter by Comparison Method	> 800 °C to 1200 °C	2.6 °C
119	THERMAL-TEMPERATURE	Thermocouple (with Controller / Indicator / Data Logger / Recorder / Scanner), Temperature Transmitter, Temperature Transducer, Temperature Gauge ,Digital Thermometer	Using S - Type Thermocouple with Indicator, 6½ Digital Multimeter by Comparison Method	> 250 °C to 600 °C	1.19 °C

* CMCs represent expanded uncertainties expressed at approximately the 95% level of confidence, using a coverage factor of k = 2.